Faculty Vitae: Roshan Yashwant Nemade

1. Education

B. E. Chemical Engineering, University of Mumbai, India, 2018
Ph.D. Chemical Engineering, University of Illinois, Chicago, IL, 2023
Post-doc Chemical Engineering, University of Illinois, Chicago, IL, 2024

2. Academic Experience

Department of Chemical Engineering, University of Illinois, Chicago, IL,
Research Assistant Professor, 2024 – Present

3. Non-Academic Experience

Grapherry, Inc., Chicago, Illinois, Co-founder,
2024-Present

Gharda Chemicals, India, In-plant trainee
06/2016 – 07/2016

4. Certifications or professional registrations

None

5. Current membership in professional organizations

None

6. Honors and awards

Postdoc Small Scholarship Award at University of Illinois Chicago, 2024.

Award for Graduate Research (AGR) at University of Illinois Chicago, 2022.

7. Service activities

1. Reviewer: Chemical Engineering Research and Design
2. Undergraduate Research Mentoring, University of Illinois Chicago
   Provided supervision and skill-building guidance for undergraduate researchers each semester and summer, including experimental safety training, data interpretation, and support for research presentations and reports.

8. Publications

1. Bhawnani, Rajan R., Michael L. Barsoum, Ankit Dhakal, Vamsi V. Gande, Roshan Y. Nemade, Prem KR Podupu, Roberto dos Reis et al. "Dynamic structural and morphological transformations in high-entropy metal-organic frameworks." *Materials Today Chemistry* 49 (2025): 103009.
2. Jaradat, Ahmad, Musawenkosi K. Ncube, Ilias Papailias, Nikhil Rai, Khagesh Kumar, Volodymyr Koverga, Roshan Y. Nemade et al. "Fast Charge-Transfer Rates in Li-CO2 Batteries with a Coupled Cation-Electron Transfer Process." *Advanced Energy Materials* (2024): 230346

3. Mukhopadhyay, Arani, Sungjoon Kim, Anish Pal, Roshan Nemade, Sreya Sarkar, Vikas Berry, and Constantine M. Megaridis. "Graphene-Enhanced Metal Condensers in Wick-Free Vapor Chambers for Thermal Management in Electronics." *In 2024 23rd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems* (ITherm), pp. 1-6. IEEE, 2024.
4. Nemade, R., Cotts, S., & Berry, V. "Graphene Fermi-Level Guided Attachment of Single Exoelectrogens and Induced Interfacial Doping", *ACS Appl. Mater. Interfaces* 2024, 16, 5, 5548–5553
5. Nemade, R., Cotts, S., & Berry, V. (2022). "Plasma treatment for enhanced microbe-electrode interfaces: A bio-electronic sink" *Journal of Power Sources*, 544, 231834. <u>Perspective article</u>
6. Dighe, A. V., Nemade, R., & Singh, M. R. (2019). "Modeling and simulation of crystallization of metal–organic frameworks." *Processes*, 7(8), 527. <u>Selected for Journal Cover</u>
7. Dr Ilias Papailias, Arash Namaeighasemi, Musawenkosi Ncube, Roshan Nemade, Dr Naveen Dandu, Nikhil Rai, Syed Ibrahim Gnani Peer Mohamed, Hessam Shahbazi, Suchit Sarin, Ahmad Jaradat, Shahriar Namvar, Pardis Seraji, Vikas Berry, Arunkumar Subramanian, Jeffrey Shield, Siamak Nejati, Dr Anh Ngo, Dr Larry Curtiss. "Pathways for Sustainable Reaction Kinetics in Li-CO2 Batteries" *Nature Sustainability*, submitted: under consideration.
8. Mr Arani Mukhopadhyay , Dr Sungjoon Kim , Mr Anish Pal , Dr Roshan Nemade , Dr Sreya Sarkar , Vikas Berry. "Synergistic integration of graphene on laser-textured copper for thermal management of high-heat-flux electronics" *Nature Electronics*, submitted: under consideration.

<u>Patents</u>

1. Berry, Vikas, Roshan Nemade, Sungjoon Kim, and Sheldon Cotts. "ELECTRON-TUNNELING OR HOPPING BASED AERIAL VIRAL DETECTOR."

<u>9. Professional Development Activities</u>

1. NSF I-Corps Technology Commercialization Training (Summer 2022)
   Participated as Co-Entrepreneurial Lead in a competitive national innovation program focused on customer discovery, value-proposition development, and commercialization pathways for engineering technologies.
2. Industrial Research Collaboration with Anheuser-Busch InBev (2020–2023)
   Designed and assembled a scale-up CVD coating system in partnership with industrial R&D staff, reinforcing skills in engineering translation and manufacturing environments.
3. Grant Development & Sponsored Research Activities
   Developed federal and state proposals as PI/Co-I/Key Personnel (DOE-LBNL, ISTC, NSF), strengthening capabilities in funding acquisition and lab sustainability.
   U.S. Patent Application# 18/743,238
4. Berry, Vikas, Roshan Nemade, and Namrita Berry. " Method and Procedure to decoupled heating and electron injection to convert carbon content into graphene and other carbon allotropes"

   Patent Application# 63/690,881

Berry, Vikas, Roshan Nemade, Kartikey Sharma. “Nanodiagnosis of Skin Cancer with graphene based device”, UIC Disclosure Ref. No. 2022-023

5. Berry, Vikas, Roshan Nemade, Kartikey Sharma. “RAPID-GR: Rapid, Large-Area Graphene Growth via Resistive Heating of Catalysts”, UIC Disclosure Ref. No. 2024-082
6. Berry, Vikas, Roshan Nemade, Kartikey Sharma. “Remote Graphene Detector”, UIC Disclosure Ref. No. 2022-009